Random Forest Hyperparameter Tuning and Performance Comparison

Model No	n_estimators	Max_depth	Min_samples_split	Min_samples_leaf	Max_features	R^2 Score
RF-01	50	None	2	1	None	0.99
RF-02	100	None	2	1	None	0.98
RF-03	200	None	2	1	None	0.98
RF-04	300	10	2	1	None	0.98
RF-05	200	15	2	1	None	0.98
RF-06	500	20	2	1	None	0.98
RF-07	300	None	4	1	None	0.98
RF-08	300	None	2	2	None	0.96
RF-09	400	None	5	2	None	0.95
RF-10	500	25	4	1	None	0.98
RF-11	600	None	2	1	None	0.98
RF-12	700	None	2	1	None	0.98
RF-13	800	30	2	1	1.0	0.98
RF-14	900	35	3	1	1.0	0.98
RF-15	1000	None	2	1	1.0	0.98

Table Random Forest Hyperparameter Tuning and performance comparison presents the performance comparison of fifteen Random Forest regression models using different hyperparameter configurations. The evaluation metric used in this study is the R^2 score, which measures the predictive accuracy of the models. Among all the models, RF-01 achieved the highest performance with an **R^2 score of 0.99**, indicating the best predictive capability